

Xenohormesis mechanisms underlying chemopreventive effects of some dietary phytochemicals.

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A wide variety of phytochemicals present in our diet, including fruits, vegetables, and spices, have been shown to possess a broad range of health-beneficial properties. The cytoprotective and restorative effects of dietary phytochemicals are likely to result from the modulation of several distinct cellular signal transduction pathways. Many dietary phytochemicals that are synthesized as secondary metabolites function as toxins, that is, "phytoalexins," and hence protect plants against insects and other damaging organisms and stresses. However, at the relatively low doses consumed by humans and other mammals, these same toxic plant-derived chemicals, as mild stressors, activate adaptive cellular response signaling, conferring stress resistance and other health benefits. This phenomenon has been referred to as xenohormesis. This review highlights the xenohormesis mechanisms underlying chemopreventive effects of some dietary chemopreventive phytochemicals, with special focus on the nuclear transcription factor erythroid 2p45 (NF-E2)-related factor 2 (Nrf2) as a key player. © 2011 New York Academy of Sciences.